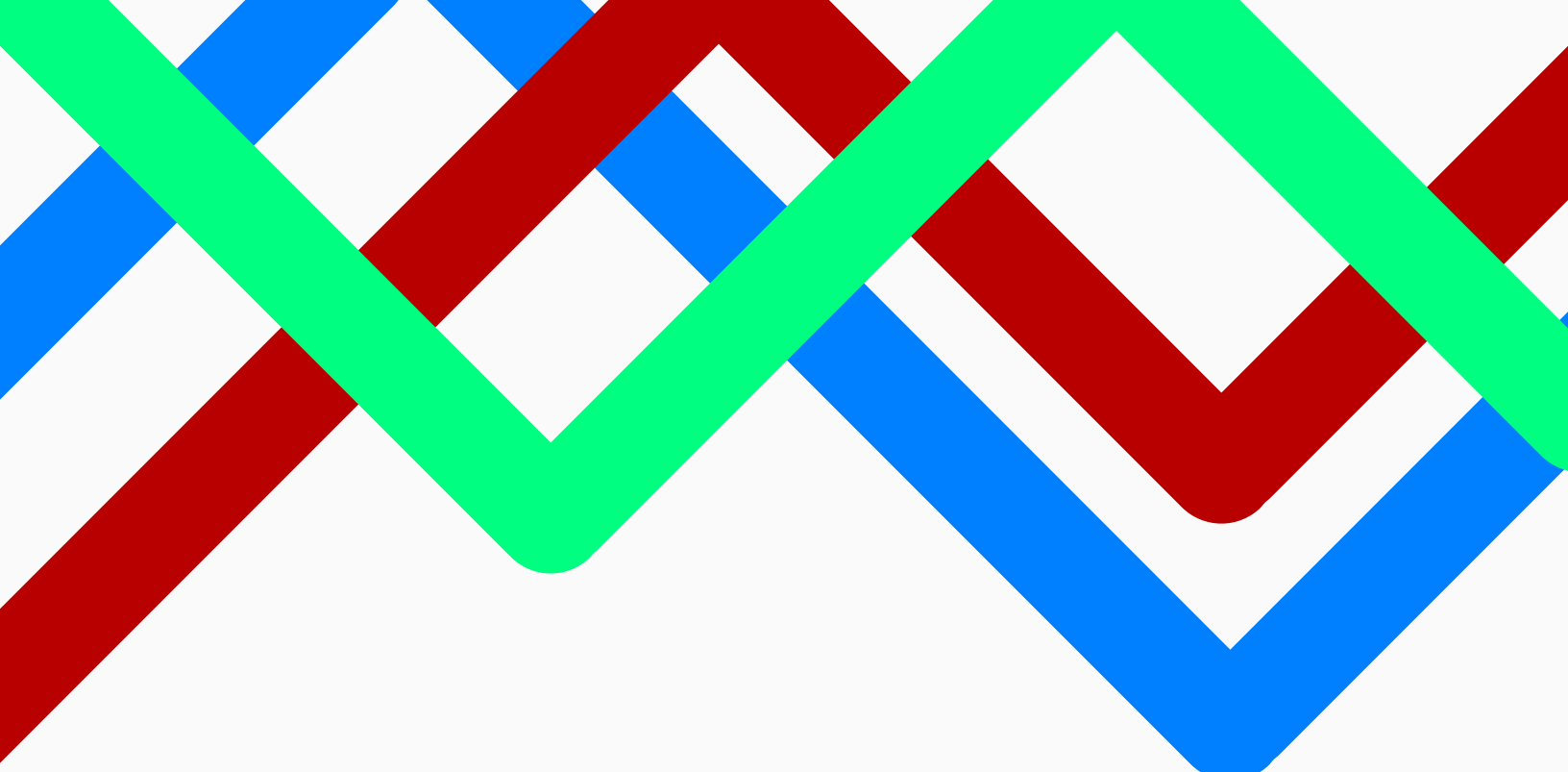




Cracker Protein Lab

Daniel Sabalakov

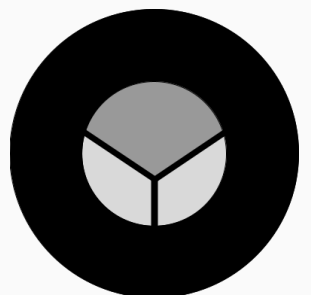
DE Intro to Biology I

October 8, 2025

“Oh My Days...” - Daniel Sabalakov

“Balright” - Turner Leduc

“Dame un grr.” - Jiles Tracz



1 Abstract

To show the denaturing of amylase through the change in pH levels and how the efficacy and efficiency of enzyme performance are scale-based.

2 Background / Introduction

3 Question / Problem

How does ***amylase***, an enzyme found in saliva, break polysaccharides into simpler carbohydrates in different PH levels?

4 Elements & Hypothesis

4.1 Hypothesis

If the pH levels of three solutions: {4, 5.5, 7}, are exposed to the enzyme *amylase* found in saliva and mixed with a cracker, then there should be a trend whereas the pH 7 solution will have the highest density of glucose, the pH 5.5 solution should have the medium density of glucose, and the pH 4 solution should have the lowest density of glucose.

4.2 Experimental Elements

- Independent Variable: PH Level of the solution
- Dependent Variable: Glucose concentration produced (in mmol/L)
- Experimental Groups:
 - Group 1: Saltine Cracker, Saliva, pH 4.0 solution
 - Group 2: Saltine Cracker, Saliva, pH 5.5 solution
- Positive Control Group:
 - Saltine Cracker, Saliva, optimal pH solution (pH 7.0)
- Negative Control Group:
 - Saltine Cracker, optimal pH solution (pH 7.0)
- Controls
 - Saliva to Solution ratio
 - Cracker to Solution ratio
 - Time allowed for enzymatic processes
 - Method of glucose measurement

5 Materials / Set up

5.1 Materials

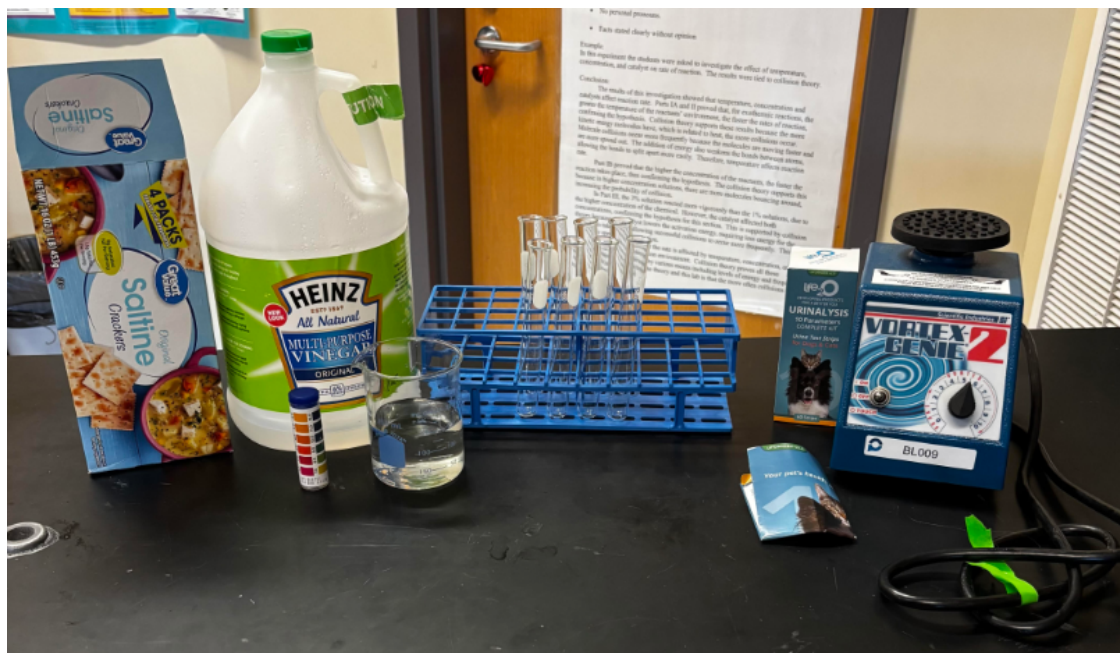
The following materials were gathered:

1. 4 saltine cracker pieces (approximately 1 gram each)
2. Roughly 100 mL of Heinz all-natural vinegar
3. pH test strips
4. Roughly 250 mL of distilled water
5. 8 15-25mL test tubes
6. Test tube rack
7. Life2O Pet Urinalysis Test Strips: Dog and Cat Urine Test Strips
8. Genie Vortex-Genie 2 Vortex Mixer SI-0236
9. Fresh Human Saliva (Roughly 20 mL total)
10. Stopwatch
11. Plastic pipettes
12. 3 stirring rods
13. Plastic gloves
14. Tape
15. Black Marker
16. 250 mL beaker

5.2 Set Up Procedure

1. All lab materials were collected and thoroughly inspected to ensure cleanliness.
2. Two sets of four test tubes were labeled with {"C","pH=7","pH=5.5","pH=4"} using tape and a marker.
3. The first set of four test tubes were filled with 10 mL of distilled water.
4. Then, according to the test tube labels, the first set had 2 mL of vinegar added incrementally and checked via a pH strip.
5. When all four test tubes of the first set were at the pH level indicated by each respective label, the tubes were put away, in a row, back onto the test tube rack

An exemplar image of an example setup without pH solutions can be found below:



6 Methods / Procedure

7 Data / Results

8 Conclusion / Discussion

9 Materials

- Vinegar (changed from citric acid for efficiency purposes)
- Water (distilled)
- Human spit/saliva (for amylase enzyme)
- Glucose Strip
- Saltine Crackers
- 4 Test tubes

10 Preparations

Be sure to have test tubes labeled with a neutral pH (this is your control group), a 6.0, a 5.0, and a 4.0. This is where each solution will be stored. Also at this time, gather 4 crackers and place them in respect to a test tube. Next to each of these crackers, be sure to have a glucose strip on hand, ready to collect when the solution interacts with the Saltines.

11 Procedure

12 Observations

13 Conclusions

14 Questions